

Impacts of manganese bio-based nanocomposites on phytochemical classification, growth and physiological responses of *Hypericum perforatum* L. shoot cultures

S. Jafarirad^{a,b,*}, M. Kosari-Nasab^{c,d,**}, R. Mohammadpour Tavana^b, S. Mahjouri^e, R. Ebadollahi^b

^a Department of Organic and Biochemistry, Faculty of Chemistry, University of Tabriz, Tabriz, Iran

^b Research Institute of Bioscience and Biotechnology, University of Tabriz, Tabriz, Iran

^c Department of Plant Biology, Faculty of Natural Sciences, University of Tabriz, Tabriz, Iran

^d Drug Applied Research Center, Tabriz University of Medical Sciences, Tabriz, Iran

^e Department of Biological Sciences, Faculty of Basic Sciences, Higher Education Institute of Rab-Rashid, Tabriz, Iran

ARTICLE INFO

Edited by Dr Muhammad Zia-ur-Rehman

Keywords:

Ecotoxicity

Nanofertilizers

Phytochemical classification

MnO₂/perlite nanocomposites

ABSTRACT

We report a new green route for preparing MnO₂/perlite nanocomposites (NCs) by leaf extract of *Hypericum perforatum*. Characterization of the physicochemical properties of the MnO₂/perlite-NCs was performed using XRD, FESEM, EDX, FT-IR, and DLS techniques. Furthermore, their effects on the phytochemical classification and growth parameters of *H. perforatum* shoot cultures were assessed. According to the FESEM image, the synthesized spherical MnO₂ nanoparticles on the sheet-like structure of nano-perlite were formed, ranging about 20–50 nm. In addition, based on the EDX spectra, the elemental analysis showed the presence of Carbon, Oxygen, Silicon, Aluminum, and Manganese elements in the as-synthesized MnO₂/perlite-NCs. Biological studies confirmed that nano-perlite and MnO₂/perlite-NCs were non-toxic to *H. perforatum* shoot cultures and showed positive effects on plant growth in specific concentrations. Overall, phytochemical classification demonstrated that the terpenoids decreased in the evaluated treatments, while hypericin and pseudohypericin were increased in some treatments (25, 50 and 150 mg/L of nano-perlite) relative to control. Metabolomics results suggested that both nano-perlite and MnO₂/perlite-NCs can be used as elicitors and new nanofertilizers for generating some secondary metabolites.

1. Introduction

Thanks to their excellent properties, nanoparticles (NPs) have widespread practical applications in various fields (Reynolds et al., 2021; Liu et al., 2020). Hence, there has been increasing interest in developing different approaches to obtain NPs with the desired shape, size, and structural properties. Currently, various physical and chemical methods are used to synthesize NPs, still most of these production approaches encounter some restrictions such as being expensive, labor-intensive, and use of toxic chemicals. However, the extracellular generation of NPs by plant-based extracts introduces a cost-effective, fast, non-toxic, and environmentally friendly production method (Noman et al., 2020). The variety of phytochemicals that exist in the

plant extracts can act as both reducing and stabilizing agents for producing various types of NPs (Ragab et al., 2020). According to reports, the biosynthesized NPs show low cytotoxicity than their counterparts fabricated using toxic reduction chemicals (Mahjouri et al., 2018). Furthermore, it has been revealed that biologically-synthesized NPs have the potential to enhance plant growth and yield of bioactive compound production (Benzon et al., 2015). Therefore, it is necessary to investigate their beneficial aspects. Recently, nanofertilizers are introduced as one of the most important biotechnological tools in modern agriculture. Nanofertilizers can significantly improve the crop production and also are ecofriendly agents compared with chemical fertilizers (El-Ghamry et al., 2018; Kalia et al., 2019). Despite promising applications of nanofertilizers for sustainable agriculture, there remain gaps in

* Corresponding author at: Department of Organic and Biochemistry, Faculty of Chemistry, University of Tabriz, Tabriz, Iran.

** Corresponding author at: Department of Plant Biology, Faculty of Natural Sciences, University of Tabriz, Tabriz, Iran.

E-mail addresses: jafarirad@tabrizu.ac.ir (S. Jafarirad), mortazakosari56@gmail.com (M. Kosari-Nasab).

<https://doi.org/10.1016/j.ecoenv.2020.111841>

Received 27 September 2020; Received in revised form 19 November 2020; Accepted 19 December 2020

Available online 30 December 2020

0147-6513/© 2020 Published by Elsevier Inc. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

our knowledge of their interactions with plants and their utilization are needed further studies.

Manganese (Mn) is generally accepted as one of the essential elements for all higher plants, which play a crucial role in plant metabolism and development. Mn supplement to the culture medium of medicinal plants improves their production yield of bioactive compounds (Ghanadnia et al., 2014; Nazari et al., 2018). Thus, Mn can be considered as an essential trace element for medicinal plants. Moreover, Mn deficiency is a common nutritional disorder in plants. Recently, the application of Mn NPs at low concentrations has been introduced as a new strategy to balance plant nutrition and enhance their growth and productivity (Liu et al., 2016; Pradhan et al., 2013). It is essential to investigate new biocompatible supports as fertilizer carriers to control the slow release of nutrients. For example various nontoxic materials such as polyvinyl alcohol and starch can be used for the slow-release formulation of nanofertilizers (Kumar et al., 2018). Besides, because of the unique characteristics such as ease of handling, cheapness, low toxicity, high stability, and non-corrosiveness, perlite can be applied as a support for the nanofertilizers (Kolvari et al., 2015). In the present study, nano-perlite was used to immobilize MnO_2 NPs on its surface. Thus, in the present study nano-perlite and MnO_2 /perlite-NCs will be evaluated as new nanofertilizers.

H. perforatum L. is an important medicinal plant with numerous pharmacological activities, including antidepressant, antiviral, and antibacterial effects. Phytochemicals such as Hyperforin and hypericin and other flavonoids and tannins are reported as the main bioactive compounds of *H. perforatum* L. (Barnes et al., 2001; Guedes et al., 2003). Therefore, these phytochemicals are essential for the biological-synthesis of NPs. In the present study, we report the biological-synthesis of MnO_2 /perlite-NCs using leaf extract of *H. perforatum*. Subsequently, as-synthesized MnO_2 /perlite-NCs and nano-perlite were evaluated using various characterization methods, including XRD, FESEM, EDX, FT-IR, and DLS. Finally, the synthesized nanostructures were investigated for their effects on phytochemical classification, growth, and physiological responses of *H. perforatum* shoot cultures.

2. Materials and methods

All the chemicals in this study were of analytical grade. Powder of perlite provided from East Azerbaijan province regions, Iran. Its chemical composition was reported previously (Rezaei et al., 2019). This mineral powder was milled (12 h under Argon gas at ball-mill ratio of 10: 1) by ball mill of Amin Asia Fanavar Pars Co (Iran, Model: NARYA-MPM 4*250). The lining of the mill was Alumina. Inert ceramic (porcelain) balls were used as grinding bodies in ball mill. The balls were made of high quality solid pressed silicate porcelain with aluminum oxide. The chemical composition for Inert Alumina Ceramic Ball was as: Al_2O_3 + SiO_2 (> 92%), Al_2O_3 (23–26%), SiO_2 (68–72%), Fe_2O_3 (0.4–0.8%) CaO (0.6–0.9%), MgO (0.6–1.2%) K_2O + Na_2O (< 3.5%), and other (< 1%). The dried plant was ground by a high speed grinder (Best Grinding Tools CO. China, 2800 W power; degree of grinding: 50–300 mesh; motor speed: 25,000 r/min). The as-synthesized samples were heated in a temperature controlled oven of Behdad-model code 3491 BCE (Behdad Co., Iran). The temperature was fixed in 70 °C. Then the samples were calcinated by a temperature controlled furnace of Kawoosh Dian Azema-model: KDA-C12 (Medical Kawoosh mega Co., Iran). The temperature was fixed at 400 °C.

2.1. Preparation of plant extract

For the preparation of plant extract, aerial parts of *H. perforatum* plants (obtained from the Botanic Garden of Tabriz University of Medical Science) were thoroughly washed with distilled water and dried at room temperature for one week. Then, the powdered plant material (10 g) was extracted with distilled water (400 ml) under constant shaking (at

25 °C for 48 h). Finally, the obtained mixture was filtered, centrifuged, and stored at 4 °C.

2.2. Extracellular biosynthesis of MnO_2 /perlite-NCs

To synthesize MnO_2 /perlite NCs, 0.5 g of manganese (IV) sulfate monohydrate (Merck) was mixed with 0.50 g of nano-perlite and 30 ml of the extract. The mixture was stirred vigorously at 25 °C for 2 h. Subsequently, the pH of the mixture was adjusted to 9 and refluxed for 6 h at 130 °C. To isolation of raw product, after centrifuging, the obtained precipitate was washed with distilled water and ethyl alcohol to remove the remained ions and other impurities in the obtained product and finally dried at 70 °C for 1 h and then calcinated at 400 °C for 4 h.

2.3. Physicochemical characterization of nano-perlite and MnO_2 /perlite-NCs

The physicochemical aspects of nano-perlite and as-synthesized MnO_2 /perlite-NCs, including crystalline structure, morphology, elemental composition, polydispersity index, zeta potential, and identification of the functional groups attached to the samples were determined by various methods. A powder diffractometer (XRD D500, Siemens Diffractometer-Germany) is configured with a graphite monochromator, $\text{CuK}\alpha$ radiation at an accelerating voltage of 40 kV/30 mA, over the range 10–90° 2 θ , step size 0.02° and 60 s count time. FESEM equipped with EDX (MIRA3 FEG-SEM, Czech Republic) was utilized to evaluate the morphological properties of samples. This microscope has a LVSTD (Low Vacuum Secondary Electron Tescan Detector) at a voltage of 30 kV. The samples were dispersed on an aluminum substrate. Then the dispersed particles were treated by gold coat onto the sample containing substrate. DLS (Nanotrac Wave, Microtrac Co. USA) was used in order to measure the hydrodynamic diameter, size distribution and zeta potential of samples. Deionized water was used as dispersant and pH adjusted on pH 5.7 and scattering angle of 90° was used for DLS evaluations at $k = 632.8$ nm. Prior to measurement, the samples were ultrasonicated in sonication bath (Bandelin Sonorex Digitec, 40KHz) was set to 25 °C for 20 min and the samples were left for 60 min before measurements. FTIR spectra were recorded using KBr pallet to record FTIR spectra on a Shimadzu model FTIR 8101 N spectrometer in the resolution of 4 cm^{-1} over the range 400–4000 cm^{-1} .

2.4. Shoot cultures and treatments

Seeds of *H. perforatum* were sterilized with ethanol (70%, v/v) for 1 min and sodium hypochlorite solution (20%, v/v) for 15 min, followed by washing with sterilized distilled water. Then, the sterilized seeds were germinated on basal Murashige and Skoog (MS) medium (3% sucrose, 0.8% agar, pH 5.6–5.8). Shoot cultures were initiated by transferring stem explants with 1 node (obtained from the 30-day-old seedlings) to the basal MS medium supplemented with the various concentrations of nano-perlite and MnO_2 /perlite-NCs including 0, 25, 50, 100, 150, 200, 250, 300, 400, and 500 mg/L. The cultures were kept at 25 °C under the photoperiod of 16 h of light and 8 h of darkness for 4 weeks.

2.5. Measurement of growth parameters

To investigate the effect of nano-perlite and MnO_2 /perlite-NCs on growth parameters of *H. perforatum* shoot cultures, including the final fresh weight (FW), dry weight (DW, dried at 35 °C for 24 h), the number of leaves, shoots, and roots as well as shoot and root length were measured.

2.6. Gas chromatography/mass spectrometry (GC-MS) analysis

To extract volatile compounds, 100 mg of fresh plant samples of each

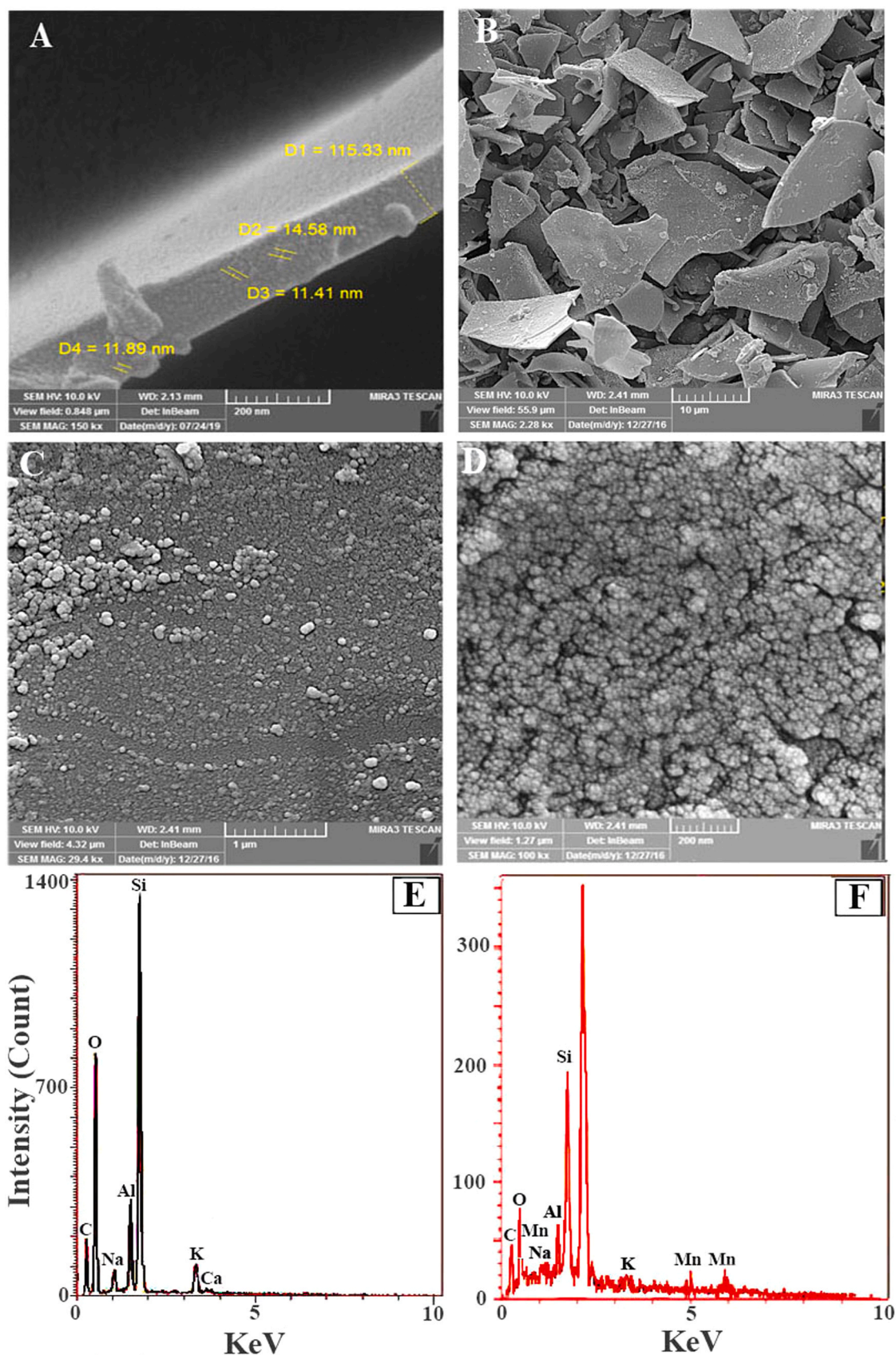


Fig. 1. FESEM images of nano-perlite (A) and MnO₂/perlite-NCs, MnO₂ nanoparticles immobilized on surface of perlite in different magnifications (B–D); EDX spectra of nano-perlite (E) and MnO₂/perlite-NCs (F).

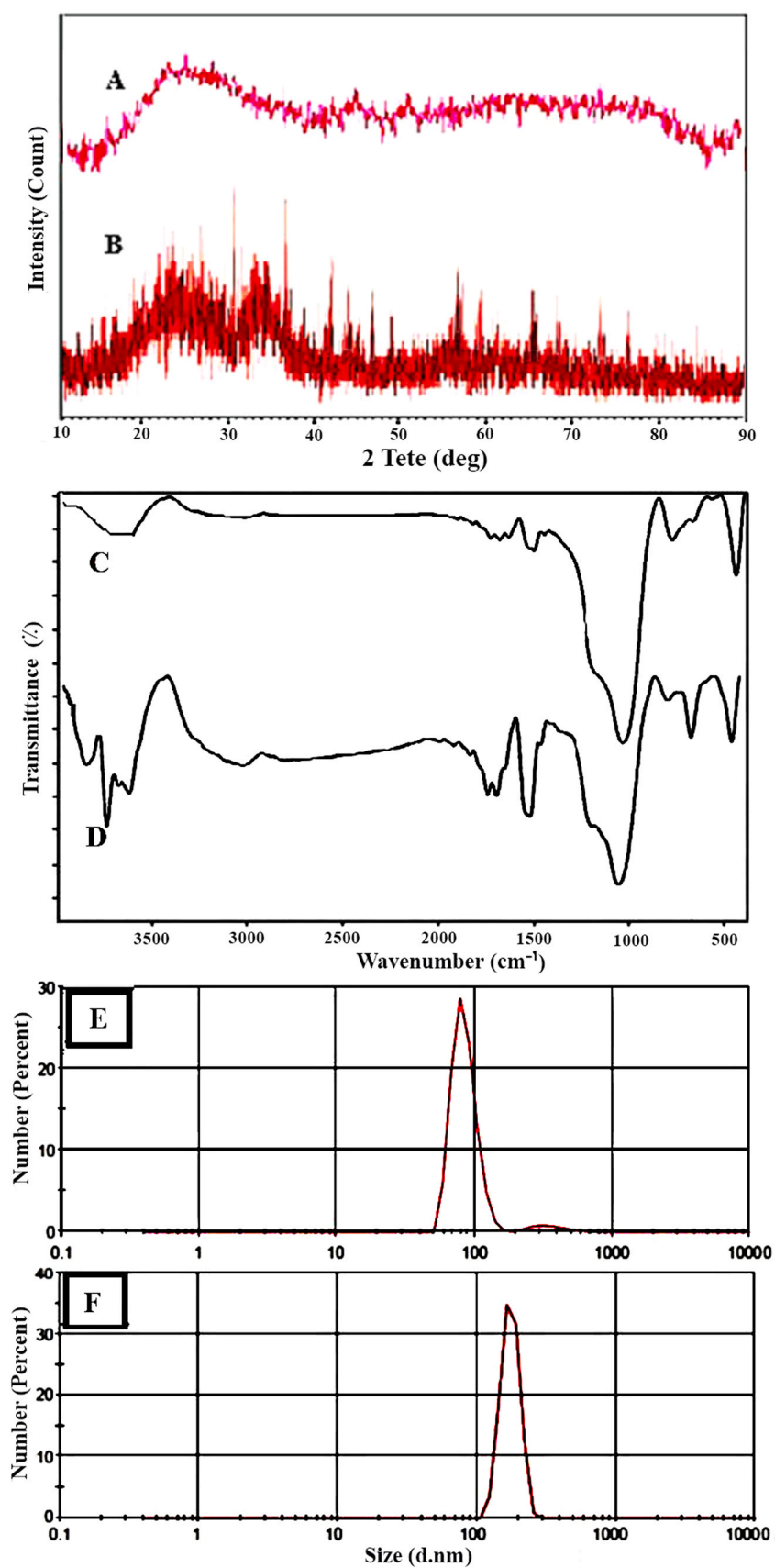


Fig. 2. XRD patterns of nano-perlite (A) and β -MnO₂/perlite-NCs (B); FT-IR spectra of nano-perlite NPs (C) and MnO₂/perlite-NCs (D); DLS patterns of nano-perlite NPs (E) and MnO₂/perlite-NCs (F).

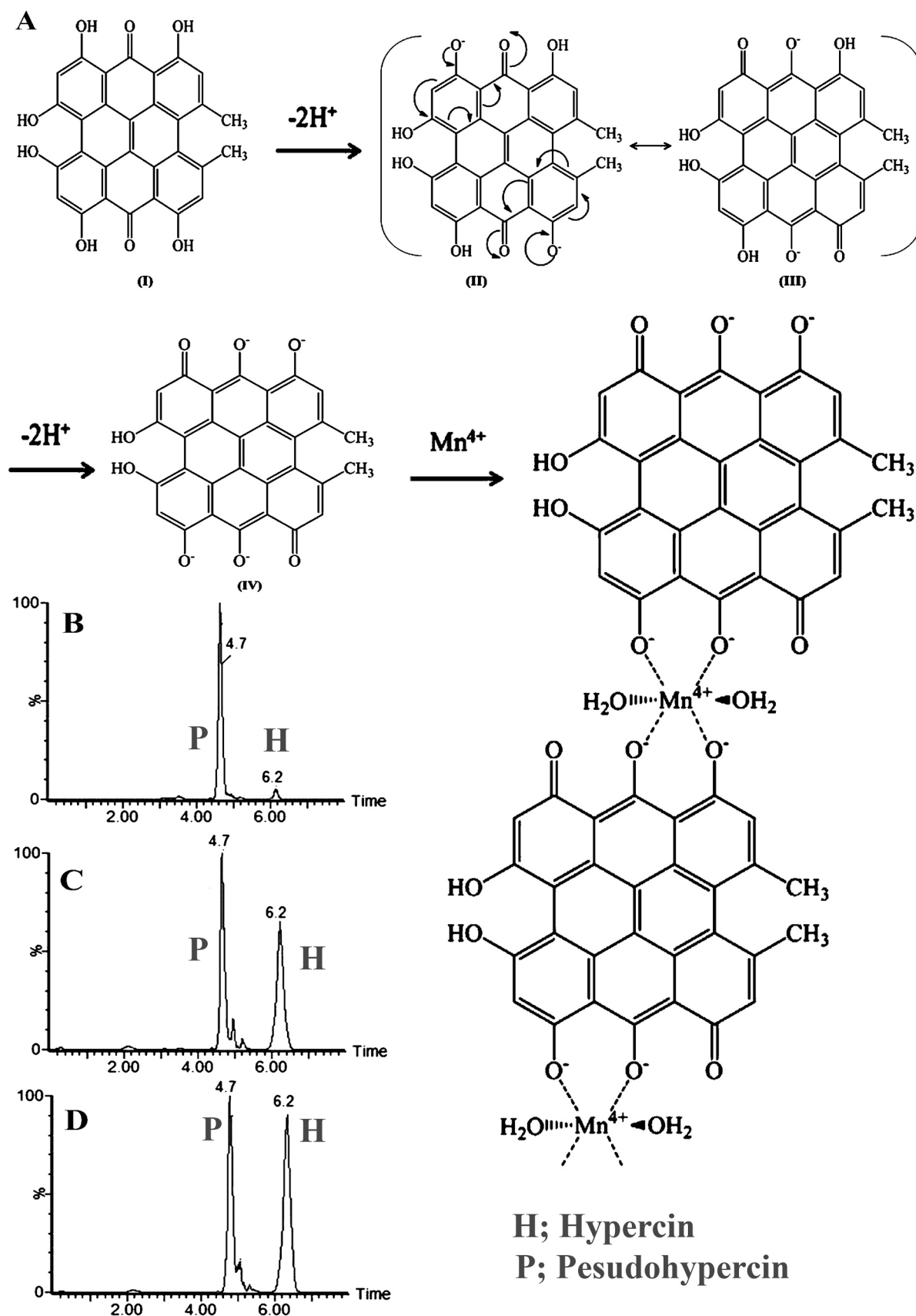


Fig. 3. The possible mechanism for the formation and stability of MnO_2 /perlite-NCs using *H. perforatum* extract, (I) in figure represents hypericin (A); LC-MS chromatograms of the extracts of control (B) 50 mg/L of nano-perlite (C), 150 mg/L of nano-perlite (D) treatments.

treatment were extracted in 6 ml of n-hexane. Obtained extracts were subjected to GC-MS analysis using Elite-I, fused silica capillary column (30 m, 0.25 mm, 0.25 μ m film thickness, composed of 100% Dimethylpolysiloxane) HP 5MS and mass spectrometer (6890 N Mass selective detector/Agilent). For GC-MS detection, an electron ionization system with ionizing energy of 70 eV was used. Helium gas (99.999%) was used as the carrier gas at a constant flow rate of 1 ml/min. The injection volume of 1 μ L, injector temperature of 150 °C, and the ion-source temperature of 280 °C were employed. The oven temperature was programmed from 50 °C (for 3 min), with an increase of 10 °C/min, to 120 °C (for 3 min), then an increase of 7 °C/min, to 220 °C (for 3 min), ending with an increase of 7 °C/min to 290 °C for 5 min, and enhanced to 300 °C. Finally, it kept there for 2 min (Guedes et al., 2003; Ebadollahi et al., 2019).

2.7. Determination of hypericin and pseudohypericin using LC-MS/MS analysis

The LC-MS/MS analysis of *H. perforatum* shoot cultures was performed via preparing the methanolic extract by dissolving dried plant samples (20 mg) in methanol (1 ml). Then, the obtained mixtures were centrifuged and filtered through a 0.45- μ m filter. The analysis was performed using Alliance separations module 2695 (Waters, Milford, MA, USA). Chromatographic separation was done using a Gemini column (50 $\times$ 2.00 mm, 5 μ m). Mass Lynx software, version 4.1, was used for instrument control and data acquisition. Mobile phases A and B were triethylammonium acetate buffer (0.01 M, pH 7) and a mixture of methanol and acetonitrile (50:50, v/v), respectively. The flow rate was set at 1.5 ml/min throughout the gradient. Approximately 10 μ L of the sample was injected. The column temperature was 40 °C. Chromatographic gradient starting with 40% A and 60% B for 0–2 min was applied. Subsequently, a linear gradient of 95% B and 5% A for 2–4 min was used and maintained for 5 min, and then the gradient returned to the initial conditions and held for 1.5 min to equilibrate the column. The total run time was 10.5 min.

The negative ion mode was selected for performing the mass analysis. The ESI negative source parameters were as follows: capillary voltage 3.5 kV, cone 60 V, extractor 2 V, RF lens 0.3 V, source temperature 110 °C, desolvation temperature 360 °C, desolvation gas (nitrogen 99.99%) flow rates 600 L/h, and cone gas flow rates 50 L/h. The analyzer settings were: unit resolution 24.0 for LM1, 14.0 for LM2 resolution, 14.0 for HM1, and 14.0 for HM2 resolution, ion energy 0.5 and 1, entrance energy 60 V and exit energy 60 V, multiplier 450 V, and Argon (99.995%, pressure 1×10^{-4} mbar) was as collision gas. Pseudohypericin and hypericin were quantified according to the peak area (RT = 4.7 and 6.2 min, respectively) compared to the standard curves. The *m/z* of both hypericin and pseudohypericin were 503 and 519, respectively.

2.8. Statistical analysis

Statistical analysis was carried out by using a one-way analysis of variance (ANOVA), and means were compared using Duncan's multiple range tests at 5% level (SPSS16). The results of this study are expressed as mean $\pm$ standard error.

3. Results and discussion

3.1. Evaluation of nano-perlite and MnO₂/perlite NCs

In this investigation, we report a biological and rapid procedure for generating MnO₂/perlite-NCs using leaf extract of *H. perforatum* as a stabilizing agent. The as-synthesized NPs were characterized by FESEM, EDX, XRD, DLS, and FT-IR techniques. The size and shape of the synthesized NPs were examined by FESEM. The FESEM images display the formation of both sheets and the spherical NPs for nano-perlite and

Table 1

Colloidal properties of nano-perlite and MnO₂/perlite-NCs.

Samples	D (nm)			PDI (μ_2/T^2) ^d	Z ^e (mV)
	XRD ^a	SEM ^b	DLS ^c		
Nano-perlite	13.72	11–14	89	0.454	30.09
MnO ₂ /perlite-NCs	50.5	20–50	270	0.061	50.60

^a Theoretical diameter calculated by Origin software based on Scherrer's equation.

^b Particle mean diameter measured by SEM.

^c Hydrodynamic diameter measured by DLS in water.

^d Polydispersity index based on DLS.

^e Zeta potential.

MnO₂, respectively. The MnO₂ NPs are placed on the surface of nano-perlite sheets for making MnO₂/perlite-NCs (Fig. 1A–D). Besides, MnO₂ NPs with different sizes from 20 to 50 nm can be observed in Fig. 1D. Besides, the EDX spectra of both nano-perlite and the as-synthesized MnO₂/perlite-NCs were represented in Fig. 1E and F, respectively. These figures confirm the presence of desire elements in both nano-perlite and MnO₂/perlite NCs. The carbon signal possibly originates from the adsorbed components of the plant extract over MnO₂/perlite NCs. The gold signal in the EDX spectrum (0.22 mv) is because of the gold-coating process in FESEM analysis (Fig. 1E). According to the EDX technique, the elemental analysis of the as-synthesized MnO₂/perlite-NCs shows the presence of Carbon, Oxygen, Silicon, Aluminum, and Manganese elements with the weight percentages of 42.70%, 34.96%, 17.33%, 2.83%, and 2.17%, respectively.

The XRD diffractograms of nano-perlite and MnO₂/perlite-NCs are shown in Fig. 2A and B. The characteristic broad reflection at $2\theta = 25^\circ$ indicates the amorphous entity of nano-perlite (Fig. 2A) (Rezaei et al., 2019). According to Fig. 2B, the new diffractions in the diffractogram of MnO₂/perlite-NCs is evident after the immobilization of MnO₂ NPs on the surface of nano-perlite (Fig. 2B). Fig. 2B displays the peak positions with 2θ value of 29.56°, 37.32°, 41.55°, 43.54°, 46.49°, 57.14°, 59.47°, 65.27°, 68.28°, 73.51° and 76.53° which are indexed as to the (110), (101), (200), (111), (210), (211), (220), (202), (310), (301) and (112) planes, respectively. Comparing the XRD patterns in Fig. 3A and B demonstrates the formation of the β -MnO₂ lattice (JCPDS no. 24-0735) as nanoparticles on the whole surface of nano-perlite (Wang et al., 2002; Yu et al., 2014; Fang et al., 2010). The average size of as-synthesized MnO₂ NPs was quantified using Debye–Scherrer equation (Behzad et al., 2020; Patterson et al., 1939). Accordingly, the average size of MnO₂ NPs was determined about 50.5 nm.

Fig. 2C and D represent the FT-IR spectra of nano-perlite and MnO₂/perlite NCs, respectively. In Fig. 2C, for nano-perlite, peaks at 453, 782, 1035, and 1350 cm^{-1} can be attributed to K–O, Ca–O, Si–O, and Al–O stretching bonds, respectively (Fig. 2C) (Rezaei et al., 2019). The typical peak at 3622, 3677, and 3739 and 3836 cm^{-1} in Fig. 2C and D correspond to the K–O–H, Ca–O–H, Si–O–H, and Al–O–H. In Fig. 2D the peak at 672 cm^{-1} is related to Mn–O bond and confirms the formation of MnO₂/perlite NCs. It has been identified that oxides and hydroxides of metal NPs have an absorption peak in the fingerprint region (below 1000 nm) attributable to inter-atomic vibrations (Jafarirad et al., 2019; Kumar et al., 2014). In Fig. 2D, also, the enhance in intensity of peak at 1051 cm^{-1} , relative to Fig. 2C, may be corresponded to the C–O stretching mode of esters of the extract. The peaks at 1696, 1742 cm^{-1} indicate the bonds C=C and C=O of esters of the extract. The peaks at 2870, 2930 and 3021 cm^{-1} can be attributed to C–H (methylene), C–H (methyl) and O–H stretching of the extract. Thus, the presence of C–O, C=C, C=O, C–H, and O–H functional groups are confirmed. These functional groups originate from some phytochemical components of *H. perforatum* leaf extract including phenols, proanthocyanidins, flavones, and flavonols (Ebadollahi et al., 2019; Jafarirad et al., 2020, 2017; Kladar et al., 2015). As will see in the next section (Possible mechanism) these components act as the reducing and stabilizing agents

Table 2Influence of different concentrations of nano-perlite and MnO₂/perlite-NCs on the growth parameters of *H. perforatum* cultures.

Treatment (mg/L)	Number (explants ⁻¹)			Length (mm)		Weight (mg)	
	Shoots	Roots	Leaves	Shoot	Root	Fresh	Dry
Control	2.38 ± 0.41 ^{bc}	0.52 ± 0.17 ^{def}	17. ± 2.89 ^b	11.30 ± 1.38	3.05 ± 1.03 ^d	21.29 ± 5.71 ^c	3.55 ± 0.95 ^{bcde}
25 perlite	2.78 ± 0.36 ^{ab}	0.10 ± 0.040 ^f	18.65 ± 2.74 ^b	10.51 ± 1.80	6.00 ± 3.34 ^{bcd}	23.78 ± 6.91 ^c	1.49 ± 0.43 ^e
50 perlite	2.78 ± 0.58 ^{ab}	1.23 ± 0.25 ^{abcde}	21.23 ± 4.27 ^{ab}	18.63 ± 2.32	7.85 ± 0.82 ^{abcd}	31.23 ± 7.61 ^{abc}	6.24 ± 1.52 ^{ab}
100 perlite	3.60 ± 0.67 ^a	1.17 ± 0.36 ^{abcde}	27.34 ± 2.54 ^a	21.86 ± 2.23	6.48 ± 1.51 ^{abcd}	41.99 ± 2.91 ^{ab}	6.00 ± 0.42 ^{ab}
150 perlite	2.40 ± 0.33 ^{bc}	1.13 ± 0.21 ^{abcde}	20.05 ± 3.11 ^b	19.49 ± 2.39	12.86 ± 2.53 ^a	28.23 ± 5.61 ^{bc}	4.79 ± 0.89 ^{abcde}
200 perlite	2.32 ± 0.33 ^{bc}	0.90 ± 0.15 ^{bcdef}	20.03 ± 1.27 ^b	16.96 ± 2.85	6.89 ± 0.89 ^{abcd}	26.17 ± 2.46 ^c	4.67 ± 0.44 ^{abcd}
250 perlite	2.28 ± 0.15 ^{bc}	2.10 ± 0.19 ^a	24.22 ± 1.47 ^{ab}	19.84 ± 2.84	10.05 ± 1.35 ^{abc}	43.95 ± 8.79 ^a	6.66 ± 1.33 ^a
300 perlite	1.90 ± 0.13 ^{bc}	1.75 ± 0.14 ^{ab}	17.67 ± 1.17 ^b	14.35 ± 3.01	7.97 ± 1.43 ^{abcd}	21.38 ± 3.62 ^c	3.89 ± 0.66 ^{bcde}
400 perlite	2.00 ± 0.14 ^{bc}	1.58 ± 0.27 ^{abc}	23.95 ± 2.42 ^{ab}	16.47 ± 2.43	5.67 ± 1.24 ^{bcd}	28.45 ± 3.92 ^{bc}	4.06 ± 0.56 ^{abcde}
500 perlite	2.00 ± 0.18 ^{bc}	1.30 ± 0.11 ^{abcde}	24.25 ± 0.87 ^{ab}	18.73 ± 2.29	6.97 ± 1.08 ^{abcd}	31.45 ± 1.30 ^{abc}	4.49 ± 0.19 ^{abcd}
25 NCs	2.06 ± 0.13 ^{bc}	0.69 ± 0.18 ^{cdef}	18.43 ± 1.46 ^b	17.23 ± 0.94	4.67 ± 0.64 ^{cd}	19.43 ± 1.11 ^c	3.89 ± 0.22 ^{bcde}
50 NCs	1.93 ± 0.14 ^{bc}	0.89 ± 0.26 ^{bcdef}	18.58 ± 1.26 ^b	19.93 ± 1.83	4.38 ± 0.81 ^{cd}	23.64 ± 1.42 ^c	3.94 ± 0.24 ^{bcde}
100 NCs	1.93 ± 0.11 ^{bc}	0.47 ± 0.18 ^{ef}	17.49 ± 0.6 ^b	14.93 ± 0.74	11.46 ± 3.53 ^{ab}	22.17 ± 1.90 ^c	3.26 ± 0.28 ^{cde}
150 NCs	1.84 ± 0.10 ^{bc}	1.77 ± 0.62 ^{ab}	17.56 ± 1.84 ^b	20.21 ± 4.24	9.48 ± 2.85 ^{abcd}	28.69 ± 6.16 ^{bc}	4.95 ± 1.07 ^{abcd}
200 NCs	1.99 ± 0.04 ^{bc}	1.41 ± 0.25 ^{abcd}	18.35 ± 1.06 ^b	18.35 ± 2.39	6.37 ± 1.64 ^{abcd}	27.41 ± 2.77 ^{bc}	4.57 ± 0.46 ^{abcd}
250 NCs	1.8 ± 0.07 ^c	1.53 ± 0.59 ^{abc}	17.05 ± 2.17 ^b	17.39 ± 3.99	11.59 ± 2.25 ^{ab}	24.68 ± 4.55 ^c	5.61 ± 1.04 ^{abc}
300 NCs	1.85 ± 0.03 ^{bc}	1.58 ± 0.27 ^{abc}	18.60 ± 1.50 ^b	14.99 ± 1.93	11.741 ± 2.36 ^{ab}	27.28 ± 6.32 ^{bc}	6.20 ± 1.44 ^{ab}
400 NCs	1.75 ± 0.06 ^c	1.55 ± 0.12 ^{abc}	20.05 ± 1.42 ^b	17.68 ± 2.47	9.31 ± 1.34 ^{abcd}	18.30 ± 2.46 ^c	2.61 ± 0.35 ^{de}
500 NCs	1.75 ± 0.10 ^c	1.70 ± 0.16 ^{ab}	19.8 ± 1.95 ^b	17.96 ± 2.56	11.70 ± 1.94 ^{ab}	18.90 ± 1.35 ^c	2.69 ± 0.20 ^{de}

Means followed by same letter are not significantly different at P < 0.05 using Duncan's test. Values represent mean ± standard error (SE).

to immobilize MnO₂ NPs on the surface of perlite successfully.

The average size, polydispersity indices (PDI) and zeta potentials of dispersed nano-perlite and MnO₂/perlite-NCs in distilled water were measured by DLS analysis. According to findings of DLS, the hydrodynamic diameter of nano-perlite and MnO₂/perlite-NCs were estimated 89 and 270 nm, respectively (Fig. 2E and F and Table 1). Given that the DLS analysis reveals the hydrodynamic size of the particles, this value may be bigger than the value obtained from FESEM and XRD (Yanhong et al., 2004). Besides, the zeta potential and PDI were determined to be +30.09 mV and 0.454 for nano-perlite and +50.6 mV and 0.0611 for MnO₂/perlite NCs. Based on the DLVO theory, the high zeta potential value puts a strong repulsive force between the particles and accordingly prohibits their aggregation (Zhang et al., 2012). However, according to DLS results the hydrodynamic diameter of MnO₂/perlite-NCs was estimated 270 nm. The higher hydrodynamic diameter shows agglomeration of the MnO₂ nanoparticles on the surface of perlite sheets to some extent. In addition, based on SEM image (Fig. 1D) some agglomeration of MnO₂ particles can be seen. In conclusion, all applied characterization methods demonstrated the successful production of nano-perlite and MnO₂/perlite NCs.

3.2. Possible mechanism for the formation of biogenic MnO₂/perlite NCs

As previously mentioned, it has been suggested that the phytochemicals of plants have a key role in the reduction and stabilization of metal ions and the formation of NPs (Nazari et al., 2020; Bhakya et al., 2016). The major constituents present in the leaf extract (Ex) of *H. perforatum* are flavonoids, naphthodianthrones, phloroglucinols, and chlorogenic acid. Hyperforin (prenylated phloroglucinol) and hypericin (a naphthodianthrone) (Barnes et al., 2001). Accordingly, the possible mechanism for the complexation of Mn⁴⁺ (which is responsible for the biosynthesis of MnO₂/perlite NC) is presented in Fig. 3A.

It can be hypothesized the lone pair electrons in the polar groups of hypericin could occupy orbital of the Mn⁴⁺ ions. Furthermore, the capping of Mn⁴⁺ with hydroxyl and carbonyl groups forms intermediate complexes (II–IV). Finally, the reaction can result in producing MnO₂ NPs on nano-perlite by calcination. Thus, it can be concluded that hypericin (along with other phytochemicals present in the extract) most probably acts as a reducing and stabilizing agent for the synthesis of MnO₂/perlite-NC. Therefore, the aqueous leaf extract of *H. perforatum* is a suitable candidate for the biosynthesis of NPs in a rapid, simple, cost-effective, and eco-friendly method (Ebadollahi et al., 2019).

3.3. Impacts of nano-perlite and MnO₂/perlite-NCs on the growth of *H. perforatum* shoot cultures

The obtained results related to plant growth aspects of *H. perforatum* shoot cultures exposed to nano-perlite and MnO₂/perlite-NCs are presented in Table 2. Shoot and leaf numbers were unaffected by both nano-perlite and MnO₂/perlite-NCs in almost all applied concentrations except for the treatment of 100 mg/L nano-perlite, which showed a significant rise as compared to the control. However, root number was affected by the NMs, with a significant increase for the treatments with 250, 300, and 400 mg/L of nano-perlite along with 150, 250, 300, 400, and 500 mg/L of MnO₂/perlite NCs. For shoot length, there was no significant difference between the control and all treated plants by nano-perlite and MnO₂/perlite-NCs. However, root length displayed a significant increase in 150 and 250 mg/L nano-perlite treated cultures as well as 100, 250, 300, and 500 mg/L MnO₂/perlite-NCs treated cultures. The fresh biomass was increased at the concentrations of 100 and 250 mg/L of nano-perlite as compared to the control. Also, the dry biomass was augmented in the culture exposed to 250 mg/L nano-perlite in comparison to the control. Whereas, for MnO₂/perlite NCs-treated cultures, no significant change was observed in the fresh and dry biomass.

The results of the present study indicated that the addition of nano-perlite and MnO₂/perlite-NCs in the culture medium not only had no adverse impact on the growth of *H. perforatum* shoot cultures, but also improved their growth in the medium supplemented with specific levels of the NMs. Mn is one of the essential elements for the growth and development of plants. According to Shebl et al. (2019), foliar application of Mn nanooxide significantly increased vegetative growth and yield characteristics, as well as the photosynthetic pigments amounts. It has been revealed that MnO_x NPs show less toxic properties in comparison to other metallic NPs and can also stimulate plant growth and crop production (Liu and Lal, 2015; de França Bettencourt et al., 2020). According to a study, Mn NPs can be considered as an effective micronutrient in comparison to MnSO₄ salt. Mn NPs were able to improve growth and photosynthesis without disturbing ROS-antioxidative balance in *Vigna radiata* plant in in vitro conditions. Most probably because of the very small sizes, these NPs can easily penetrate into plants and translocate well inside them. So, they could be applied as a plant micronutrient for increasing growth and photosynthesis (Pradhan et al., 2013). In conclusion, because of the low toxicity of both perlite and Mn NPs, MnO₂/perlite-NCs can be considered as the potential nano-fertilizers for improving medicinal plant growth and development.

Table 3Phytochemical classification of the volatile compounds obtained from the shoot cultures of *H. perforatum* treated with nano-perlite and MnO₂/perlite-NCs.

Raw	Compounds	Control	Perlite 50 mg/L	Perlite 100 mg/L	Perlite 150 mg/L	Perlite 200 mg/L	Perlite 250 mg/L	Perlite 300 mg/L	Perlite 400 mg/L
1	alpha-Pinene	+	–	–	–	–	–	–	–
2	beta-Pinene	+	–	–	–	–	–	–	–
3	beta-Myrcene	+	–	–	–	–	–	–	–
4	Carvacrol	+	–	+	–	–	–	–	–
5	Thymol	+	–	+	–	–	–	–	–
6	Phenol, p-tert-butyl	–	–	–	–	–	–	–	–
7	2-Ethyl-4,5-dimethylphenol	+	–	–	–	–	–	–	–
8	Bicyclo[4.4.0]dec-1-ene, 2-isopropyl-5-methyl-9-methylene	+	–	–	–	–	–	–	–
9	Germacrene D	+	–	+	–	–	–	–	–
10	Phenol, 2,6-bis(1,1-dimethylethyl)-4-methyl	–	–	–	–	–	–	+	–
11	alpha-Copaene	+	–	–	–	–	–	–	–
12	Thiophene, 2-butyl	–	–	–	–	–	–	–	–
13	1,2-Benzenediol, 3,5-bis(1,1-dimethylethyl)	–	–	–	–	–	–	–	–
14	Nonadecane	–	–	–	–	–	–	–	–
15	Tetradecanoic acid	–	–	–	–	–	–	–	–
16	Trimecaine	–	–	–	–	–	–	+	–
17	Hexadecanoic acid, methyl ester	–	–	–	+	–	–	–	–
18	3-Decen-5-one, 2-methyl	–	–	–	–	–	–	+	–
19	1-Nonadecene	–	–	–	–	–	–	+	–
20	Heneicosane	–	–	–	–	–	–	–	–
21	9-Octadecenoic acid (Z)-, methyl ester	–	–	–	+	–	–	–	–
22	Heneicosane, 3-methyl	+	–	+	+	–	–	–	–
23	14-. BETA.-H-PREGNA	–	–	–	–	–	–	+	–
24	Oxalic acid, cyclohexylmethyl octyl ester	–	–	–	–	–	–	–	–
25	n-Tetracosane	–	–	–	+	–	–	–	–
26	Pentacosane	–	–	–	–	–	–	–	–
27	4,6-cholestadien-3.beta.-ol	–	–	–	–	–	–	–	–
28	Hexacosene	–	–	–	–	–	–	–	–
29	Nonacosane	–	–	–	–	–	–	–	–
30	Carvone acetate	–	+	–	–	–	–	–	+
31	N-Methyl-1-adamantaneacetamide	–	–	–	–	–	–	+	–
32	n-Octacosane	–	–	–	+	–	–	–	–
33	5,14,23-Octadecatrien-14,15-diol	–	–	–	–	–	–	–	–
34	2,4-Cyclohexadien-1-one,3,5-bis(1,1-dimethylethyl)-4-hydroxy	–	–	–	–	–	–	+	–
35	2-(Acetoxymethyl)-3-(methoxycarbonyl)biphenylene	+	+	+	–	–	–	+	–

^{RT} Retention time.

Currently, there are few reports on the potential use of perlite and Mn-based NPs as nanofertilizer and it should be addressed in future studies.

3.4. Volatile compounds profiles of *H. perforatum* shoot cultures in response to nano-perlite and MnO₂/perlite-NCs

GC-MS analysis identified the variability of the volatile compounds of untreated and treated *H. perforatum* shoot cultures. Table 3 presents the relative abundance and phytochemical classification of the identified compounds. Comparison of the GC-MS results showed that the qualitative profile was considerably different in control and treated cultures with nano-perlite and MnO₂/perlite-NCs. Monoterpenes and sesquiterpenes, were the major components of the extract obtained from the control cultures. Monoterpenes (hydrocarbon monoterpenes and oxygenated monoterpenes) predominated the compounds, followed by hydrocarbon sesquiterpenes. Germacrene D and α -pinene were the major compounds in terpene groups. In accordance with our results, α -pinene and germacrene D were determined as the major constituents of the essential oils of *H. perforatum* from different geographical origins (Guedes et al., 2012). Also, it has been reported that in vitro shoots cultures showed sesquiterpene hydrocarbons as the main group (40.3%) and *n*-nonane as abundant compound (24.2%) (Bertoli et al., 2011).

Some phytochemical compounds such as α -pinene, β -pinene, β -myrcene, α -copaene, and Bicyclo [4.4.0] dec-1-ene, 2-isopropyl-5-methyl-9-methylene were detected in the control extract, but they

were absent from the extract obtained from treated cultures. None of the treated cultures showed hydrocarbon monoterpenes in their extracts except for the treatments of 100 mg/L of nano-perlite and 50 mg/L of MnO₂/perlite-NCs that indicated oxygenated monoterpenes. Also, hydrocarbon sesquiterpenes were observed in the cultures exposed to 100 mg/L of nano-perlite. In general, terpenoids were emitted in most of the cultures treated with nano-perlite and MnO₂/perlite NCs, or they existed in lower numbers in comparison to the control. In contrast to our results, increased amounts of monoterpenes have been identified in the essential oil of *Tanacetum parthenium* at high Mn concentrations (Farzadfar et al., 2017). It was also reported that Mn considerably improved monoterpene production in *Cuminum cyminum* flowers (Ghannadnia et al., 2014). Correspondingly, elevation in the expression levels of the genes required for the terpenoid production has been revealed in *Mentha aquatica* upon the application of Mn (Nazari et al., 2017). Information about the impact of Mn on the biosynthesis of terpenoids is restricted, and more research is required.

Our results showed the positive impact of nano-perlite and MnO₂/perlite-NCs supply on the production of some metabolites like aliphatic hydrocarbons and phenolic, in comparison with the control. Aliphatic hydrocarbons were mainly produced in the cultures exposed to 150 mg/L of nano-perlite and 250 mg/L of MnO₂/perlite-NCs. Most abundant phenolics were identified in the cultures supplemented with 300 mg/L of nano-perlite. Better bioactivity of the elicitor-treated plants compared to the untreated ones because of the variation in the phytochemical profile of *H. perforatum* under the influence of different elicitors has been

Perlite 500 mg/L	NCs 50 mg/L	NCs 100 mg/L	NCs 150 mg/L	NCs 200 mg/L	NCs 250 mg/L	NCs 300 mg/L	NCs 400 mg/L	NCs 500 mg/L	Classification	RT
-	-	-	-	-	-	-	-	-	Monoterpene	6.14
-	-	-	-	-	-	-	-	-	Monoterpene	7.10
-	-	-	-	-	-	-	-	-	Monoterpene	7.39
-	+	-	-	-	-	-	-	-	Monoterpene-oxygenated	12.39
-	-	-	-	-	-	-	-	-	Monoterpene-oxygenated	12.86
-	+	-	-	-	-	-	-	-	phenolic	13.79
-	-	-	-	-	-	-	-	-	Phenolic	14.35
-	-	-	-	-	-	-	-	-	Sesquiterpen	17.85
-	-	-	-	-	-	-	-	-	Sesquiterpen	17.39
-	-	-	-	-	-	-	-	-	Phenolic	18.6
-	-	-	-	-	-	-	-	-	Sesquiterpen	18.96
-	-	-	-	-	-	+	-	-	Unknown	23.12
-	-	-	-	-	+	-	-	-	Phenolic	23.91
-	-	-	-	-	+	-	-	-	Aliphatic	24.09
-	-	-	-	-	+	-	-	-	Fatty acid	24.87
-	-	-	-	-	-	-	-	-	unknown	26.78
-	-	-	-	-	-	-	-	-	Fatty acid	27.23
-	-	-	-	-	-	-	-	-	keton	28.61
-	-	-	-	-	-	-	-	-	Aliphatic	28.78
-	-	-	-	-	+	-	-	-	Aliphatic	29.71
-	-	-	-	-	-	-	-	-	Fatty acid	29.82
-	+	-	-	-	-	-	+	-	Aliphatic	30.39
-	+	-	-	-	-	-	-	-	Steroid	34.34
-	-	-	-	-	-	-	-	-	Ester	34.38
-	-	-	-	-	-	-	-	-	Aliphatic	35.18
-	-	-	-	-	+	-	-	-	Aliphatic	35.34
-	-	-	-	-	+	-	-	-	Steroid	38.19
-	-	-	-	-	+	-	-	-	Aliphatic	38.74
-	-	-	-	-	+	-	-	-	Aliphatic	38.87
+	+	+	+	+	-	+	+	+	Monoterpene-oxygenated	39.85
-	-	-	-	-	-	-	-	-	Amide	40.24
-	-	-	-	-	-	-	-	-	Aliphatic	40.67
-	-	-	-	-	+	-	-	-	Alcohol	40.87
-	-	-	-	-	-	-	-	-	phenolic	41.38
+	+	-	+	-	-	-	+	+	Aromatic	44.96

reported (Shakya et al., 2019). Therefore, the impact of elicitors like NMs on the pharmaceutical properties of *H. perforatum* cultures should be investigated in further studies in more detail.

3.5. Effect of nano-perlite and MnO₂/perlite-NCs in producing of hypericin and Pseudohypericin

Due to the fact that hypericin is present in the closeness of the dark nodules, most of the studies about the elicitation of this compound have been done in the seedlings or shoot cultures. Therefore, in this study impact of nano-perlite and MnO₂/perlite-NCs in producing of both hypericin and pseudohypericin was investigated in shoot cultures of *H. perforatum*. Both concentrations of 50 and 150 mg/L of nano-perlite resulted in enhancing Hypericin production. Nevertheless, higher concentrations of nano-perlite (200–500 mg/L) led to the reduction or inhibition of hypericin biosynthesis (Table 4, Fig. 3B–D). Similarly, low concentrations of nano-perlite (25, 50, and 150 mg/L) were more effective than higher concentrations (250–500 mg/L) in the induction of pseudohypericin. Moreover, the addition of MnO₂/perlite-NCs resulted in lessening of the amounts of both hypericin and pseudohypericin (except for the treatment of 250 mg/L) in *H. perforatum* shoot cultures as compared to the control (Table 4).

Besides several well-known biotic and abiotic elicitors, NPs have been recently announced as a new class of elicitors for the inducing the amounts of hypericin in *H. perforatum* in vitro cultures. For example, it has been reported that the exposure of *H. perforatum* cell cultures to

some metal oxides such as zinc and iron oxide nanoparticles increased the generation of hypericin and hyperforin (Sharafi et al., 2013). Our results also revealed the positive effect of low concentrations of nano-perlite as an elicitor to enhance hypericin and pseudohypericin production. However, Mn negatively influenced their accumulation, which needs more investigations.

4. Conclusions

The present investigation introduced an eco-friendly, fast and low-cost approach to synthesize MnO₂/perlite-NCs using the extract of *H. perforatum* plant. Stabilization was accomplished possibly because of the presence of certain phytochemicals such as hypericin in the plant. This study confirmed that nano-perlite and MnO₂/perlite-NCs were nontoxic to *H. perforatum* shoot cultures and even showed positive effects on plant growth. GC-MS based metabolite analysis demonstrated that the exposure of the cultures to nano-perlite and MnO₂/perlite-NCs can result in considerable changes in bioactive compounds accumulation. The production of certain phytochemicals consisted of phenolics and aliphatic hydrocarbons were upregulated in the exposure of different concentrations of nano-perlite and MnO₂/perlite-NCs in comparison to the control. Hypericin and pseudohypericin production was increased in the shoot cultures treated with low concentrations of nano-perlite. However, higher concentrations of nano-perlite and most of the applied concentrations of MnO₂/perlite-NCs resulted in reducing deals of hypericin and pseudohypericin. Indeed, available evidence indicates

Table 4

Comparison of hypericin and pseudohypericin contents in cultures of *H. perforatum* treated with different concentrations of nano-perlite and MnO₂/perlite-NCs.

Treatment (mg/L)	Hypericin (µg/g DW)	Pseudohypericin (µg/g DW)
Control	91.66	339.38
25 perlite	107.23	452.18
50 perlite	308.63	938.08
100 perlite	94.62	310.42
150 perlite	251.81	627.23
200 perlite	8.92	357.25
250 perlite	–	14.62
300 perlite	–	–
400 perlite	46.34	106.66
500 perlite	–	–
25 NCs	–	–
50 NCs	–	189.71
100 NCs	–	212.43
150 NCs	6.21	156.10
200 NCs	–	–
250 NCs	–	461.71
300 NCs	–	1.85
400 NCs	–	150.94
500 NCs	17.00	–

that nano-perlite and MnO₂/perlite-NCs can be suitable candidates in plant fertilization and production of qualified medicinal plants when used at specific concentrations.

CRediT authorship contributions statement

Saeed Jafarirad and Morteza Kosari-Nasab conceived and designed the study. Saeed Jafarirad, Reyhaneh Mohammadpour Tavana, Morteza Kosari-Nasab, Sepideh Mahjouri and Roghaieh Ebadollahi performed the study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors wish to thank the University of Tabriz for financial supports.

Conflicts of interest

We as the authors of the presented paper state that there are not any potential conflicts of interest.

References

Behzad, F., Jafarirad, S., Samadi, A., Barzegar, A., 2020. A systematic investigation on spectroscopic, conformational, and interactional properties of polypeptide/nanomaterial complex: effects of biobased synthesized maghemite nanocomposites on human serum albumin. *Soft Mater.* 18, 471–486 doi.org/10.1080/1539445X.2020.1716800.

Benzon, H.R.L., Rubenecia, M.R.U., Ultra, V.U., Chul Lee, S., 2015. Nano-fertilizer affects the growth, development, and chemical properties of rice. *Int. J. Agron. Agric. Res.* 7, 105–117.

Bertoli, A., Cirak, C., Silva, J., 2011. Hypericum species as sources of valuable essential oils. *Med. Aromat. Plant Sci. Biotechnol.* 5, 29–47.

Bhakya, S., Muthukrishnan, S., Sukumaran, M., Muthukumar, M., 2016. Biogenic synthesis of silver nanoparticles and their antioxidant and antibacterial activity. *Appl. Nanosci.* 6, 755–766.

Ebadollahi, R., Jafarirad, S., Kosari-Nasab, M., Mahjouri, S., 2019. Effect of explant source, perlite nanoparticles and TiO₂/perlite nanocomposites on phytochemical composition of metabolites in callus cultures of *Hypericum perforatum*. *Sci. Rep.* 9, 1–15.

El-Ghamry, A., Mosa, A.A., Alshaal, T., El-Ramady, H., 2018. Nanofertilizers vs. biofertilizers: new insights. *Environ. Biodivers. Soil Secur.* 2, 51–72.

Fang, S., 2010. Adsorption and decomposition of triclosan by activated carbons, manganese oxides, and their composite materials from aqueous solution, University of South Australia.

Farzadfar, S., Zarinkamar, F., Hojati, M., 2017. Magnesium and manganese affect photosynthesis, essential oil composition and phenolic compounds of *Tanacetum parthenium*. *Plant Physiol. Biochem.* 112, 207–217.

de França Bettencourt, G.M., Degenhardt, J., Torres, L.A.Z., de Andrade Tanobe, V.O., Soccol, C.R., 2020. Green biosynthesis of single and bimetallic nanoparticles of iron and manganese using bacterial auxin complex to act as plant bio-fertilizer. *Biocatal. Agric. Biotechnol.* 30, 101822.

Ghannadnia, M., Haddad, R., Zarinkamar, F., Sharifi, M., 2014. Manganese treatment effects on terpene compounds of *Cuminum cyminum* flowers. *Ind. Crops Prod.* 53, 65–70.

Guedes, A.P., Amorim, L.R., Vicente, A.M.S., 2003. Essential oils from plants and in vitro shoots of *Hypericum androsaemum* L. *J. Agric. Food Chem.* 51, 1399–1404.

Guedes, A.P., Franklin, G., Fernandes-Ferreira, M., 2012. *Hypericum* sp.: essential oil composition and biological activities. *Phytochem. Rev.* 11, 127–152.

Jafarirad, S., Kordi, M., Kosari-Nasab, M., 2017. Extracellular one-pot synthesis of nanosilver using *Hyssopus officinalis* L.: a biophysical approach on bioconstituent-Ag⁺ interactions. *Inorg. Nano-Met. Chem.* 47, 632–638.

Jafarirad, S., Salmasi, M., Divband, B., Sarabchi, M., 2019. Systematic study of Nd³⁺ on structural properties of ZnO nanocomposite for biomedical applications; in-vitro biocompatibility, bioactivity, photoluminescence and antioxidant properties. *J. Rare Earths* 37, 508–514.

Jafarirad, S., Taghizadeh, P.M., Divband, B., 2020. Biosynthesis, characterization and structural properties of a novel kind of Ag/ZnO nanocomposites in order to increase its biocompatibility across human A549 cell line. *Bionanoscience* 10, 42–53.

Kalia, A., Kaur, H., 2019. Nano-biofertilizers: harnessing dual benefits of nano-nutrient and bio-fertilizers for enhanced nutrient use efficiency and sustainable productivity. In: Pudake, R., Chauhan, N., Kole, C. (Eds.), *Nanoscience for Sustainable Agriculture*. Springer, Cham. https://doi.org/10.1007/978-3-319-97852-9_3.

Kladar, N., Srdrenović, B., Grujić, N., Rat, M., Gavarić, N., Anačko, G., Božin, B., 2015. St. John's Wort (*Hypericum* Spp.) - relation between the biological source and medical properties. In: *Hypericum: Botanical Sources, Medical Properties and Health Effects*, pp. 53–80.

Kolvári, E., Koukabi, N., Hosseini, M.M., Khandani, Z., 2015. Perlite: an inexpensive natural support for heterogenization of HBF 4. *RSC Adv.* 5, 36828–36836.

Kumar, R., Ashfaq, M., Verma, N., 2018. Synthesis of novel PVA-starch formulation-supported Cu-Zn nanoparticle carrying carbon nanofibers as a nanofertilizer: controlled release of micronutrients. *J. Mater. Sci.* 53, 7150–7164.

Kumar, B., Karikattb, S., Krishna, S.H., Udayashankarab, T.H., Shivaprasada, K.H., Nagabhushanac, B.M., 2014. Synthesis, characterization of nano MnO₂ and its adsorption characteristics over an azo dye. *J. Mater. Sci.* 2, 27–31.

Liu, R., Lal, R., 2015. Potentials of engineered nanoparticles as fertilizers for increasing agronomic productions. *Sci. Total Environ.* 514, 131–139.

Liu, Y., Pan, B., Li, H., Lang, D., Zhao, Q., Zhang, D., Wu, M., Steinberg, S.E.W., Xing, B., 2020. Can the properties of engineered nanoparticles be indicative of their functions and effects in plants? *Ecotoxicol. Environ. Saf.* 205, 111128.

Mahjouri, S., Movafeghi, A., Divband, B., Kosari-Nasab, M., 2018. Toxicity impacts of chemically and biologically synthesized CuO nanoparticles on cell suspension cultures of *Nicotiana tabacum*. *Plant Cell Tissue Organ Cult.* 135, 223–234.

Nazari, F., Jafarirad, Movafeghi, A., Kosari-Nasab, M., Mohajjel Kazumi, E., 2020. Toxicity of microwave-synthesized silver-reduced graphene oxide nanocomposites to the microalgae *Chlorella vulgaris*: comparison with the hydrothermal method synthesized counterparts. *J. Environ. Sci. Health Part A* 55, 639–649.

Nazari, M., Zarinkamar, F., Soltani, B.M., 2017. Physiological, biochemical and molecular responses of *Mentha aquatica* L. to manganese. *Plant Physiol. Biochem.* 120, 202–212.

Nazari, M., Zarinkamar, F., Soltani, B.M., Niknam, V., 2018. Manganese-induced changes in glandular trichomes density and essential oils production of *Mentha aquatica* L. at different growth stages. *J. Trace Elem. Med. Biol.* 50, 57–66.

Noman, M., Shahid, S., Ahmed, T., Tahir, M., Naqqash, T., Muhammad, S., Song, F., Arslan Abid, H.M., Aslam, Z., 2020. Green copper nanoparticles from a native *Klebsiella pneumoniae* strain alleviated oxidative stress impairment of wheat plants by reducing the chromium bioavailability and increasing the growth. *Ecotoxicol. Environ. Saf.* 192, 110303.

Patterson, A., 1939. The Scherrer formula for X-ray particle size determination. *Phys. Rev.* 56, 978–982.

Pradhan, S., Patra, P., Das, S., Chandra, S., Mitra, Sh, Dey, K.K., Akbar, Sh, Palite, P., Goswami, A., 2013. Photochemical modulation of bio-safe manganese nanoparticles on *Vigna radiata*: a detailed molecular, biochemical, and biophysical study. *Environ. Sci. Technol.* 47, 13122–13131.

Ragab, G.A., Saad-Allah, K.M., 2020. Green synthesis of sulfur nanoparticles using *Ocimum basilicum* leaves and its prospective effect on manganese-stressed *Helianthus annuus* (L.) seedlings. *Ecotoxicol. Environ. Saf.* 191, 110242.

Reynolds, A., Giltrap, M., Chambers, G., 2021. Acute growth inhibition & toxicity analysis of nano-polystyrene spheres on *Raphidocelis subcapitata*. *Ecotoxicol. Environ. Saf.* 207, 111153.

Rezaei, Z., Jafarirad, S., Kosari-Nasab, M., 2019. Modulation of secondary metabolite profiles by biologically synthesized MgO/perlite nanocomposites in *Melissa officinalis* plant organ cultures. *J. Hazard. Mater.* 3800, 120878–12089.

Shakya, P., Marslin, G., Siram, K., Beerhues, L., Franklin, G., 2019. Elicitation as a tool to improve the profiles of high-value secondary metabolites and pharmacological properties of *Hypericum perforatum*. *J. Pharm. Pharmacol.* 71, 70–82.

- Sharafi, E., Nekoei, S.M., Fotokian, M.H., Davoodi, D., Hadavand, Mirzaei, H., Hasanloo, T., 2013. Improvement of hypericin and hyperforin production using zinc and iron nano-oxides as elicitors in cell suspension culture of St John's wort (*Hypericum perforatum* L.). *J. Med. Plants By-prod.* 2, 177–184.
- Shebl, A., Hassan, A.A., Salama, D.M., El-Aziz, A., Abd Elwahed, M.S., 2019. Green synthesis of nanofertilizers and their application as a foliar for *Cucurbita pepo* L. *J. Nanomater.* 11, 2019.
- Wang, X., Li, Y., 2002. Selected-control hydrothermal synthesis of α - and β -MnO₂ single crystal nanowires. *J. Am. Chem. Soc.* 124, 2880–2881.
- Yanhong, L., Dejun, W., Qidong, Zh, Min, Y., Qinglin, Zh, 2004. A study of quantum confinement properties of photogenerated charges in ZnO nanoparticles by surface photovoltage spectroscopy. *J. Phys. Chem. B* 108, 3202–3206.
- Yu, S., Liu, R., Yang, W., Han, K., Wang, Zh, Zhu, H., 2014. Synthesis and electrocatalytic performance of MnO₂-promoted Ag@ Pt/MWCNT electrocatalysts for oxygen reduction reaction. *J. Mater. Chem. A* 2, 5371–5378.
- Zhang, C., Chung, J.W., Priestley, R.D., 2012. Dialysis nanoprecipitation of polystyrene nanoparticles. *Macromol. Rapid Commun.* 33, 1798–1803.